Module 1 Digital transformation with google cloud

Paardigm Shift: fundamental and irreversible change in the way human work and engage

Digital transformation

When an organization uses new digital technologies, such as public, private, and hybrid cloud platforms to create or modify business processes, culture, and customer experiences to meet the needs of changing business and market dynamics.

Cloud is the metaphor for the network of data centers.

Fundamental of Cloud concept

- **On-prem -** Everything physical, managed by owner
- **Private, Hybrid, Multi cloud**
- **Private cloud -** private environment (like on prem but run in local environment)
    - **Benefits** - Self-service, scalability, Elasticity
- **Hybrid -** combination of different environment (private & public cloud)
- **Multi cloud - more than one public cloud operators like GCP+AWS+AZURE**

Why use Hybrid or Multi Cloud?

- Access to latest tech
- Modernizing tech
- Improved ROI
- Flexibility
- Avoid vendor lock-in concerns
- Improve reliability and resiliency
- Maintain regulatory compliance
- Running main apps or software on-prem cloud

Benefits of cloud computing

Scalability, flexibility, agility, strategic value, security, cost effectiveness

Cloud Eras

- VM cloud era
- Infrastructure cloud era
- Digital transformation cloud era

Biggest challenge needs to accelerate digital transformation

- Data, infrastructure, hybrid workplace, security, sustainability

Capabilities of Google transformation cloud

- Data, open infrastructure, collaboration, trust, sustainable energy

Data Cloud: unified solution to manage data across the entire data lifecycle

Sustainable technology: save co2 emission, carbon free, reduced carbon footprint

Google cloud adoption framework serves as map

1. Short term tactical objectives
2. Mid-term strategic objectives
3. Long-term transformational objectives

Total cost of ownership (TCO)
CapEx vs OpEx : CapEx is an investment in the future, while **OpEx is the cost of running today.**

1. CapEx (Capital Expenditure)

CapEx refers to major, upfront investments in physical assets that a company will own and use for several years.

- Vibe: "Buying it once."
- Financial Impact: You pay a huge amount of money at the start. You don't get to deduct the whole cost from your taxes right away; instead, the cost is "depreciated" (spread out) over the many years you own the asset.

2. OpEx (Operating Expenditure)

OpEx refers to the day-to-day costs of keeping a business running. These are recurring expenses that provide immediate benefits.

- Vibe: "Paying as you go."
- Financial Impact: You pay for what you use, usually monthly or yearly. These costs are fully tax-deductible in the same year you pay for them.

Infrastructure procurement changes in the cloud

Any employee can create resources in seconds on infrastructure owned and managed by a cloud provider.

Support of network for digital transformation

- Fiber optic network - optical fibers transport light

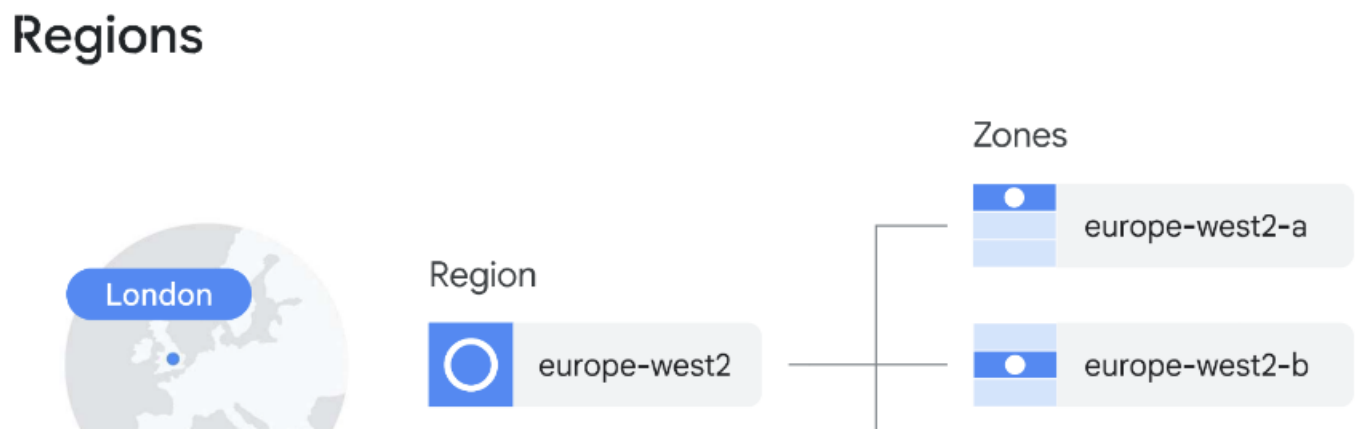
- ISP (internet service provider) - Verizon, Vodafone, Softbank, etc. ,365x24x7
- IP address (internet protocol): identify a network or location of particular device on a network
- Domain name: www.example.com
- DNS - Domain Name System: DNS server is a database of domain names mapped to IP address used by computers to communicate with each other.

Network Performance

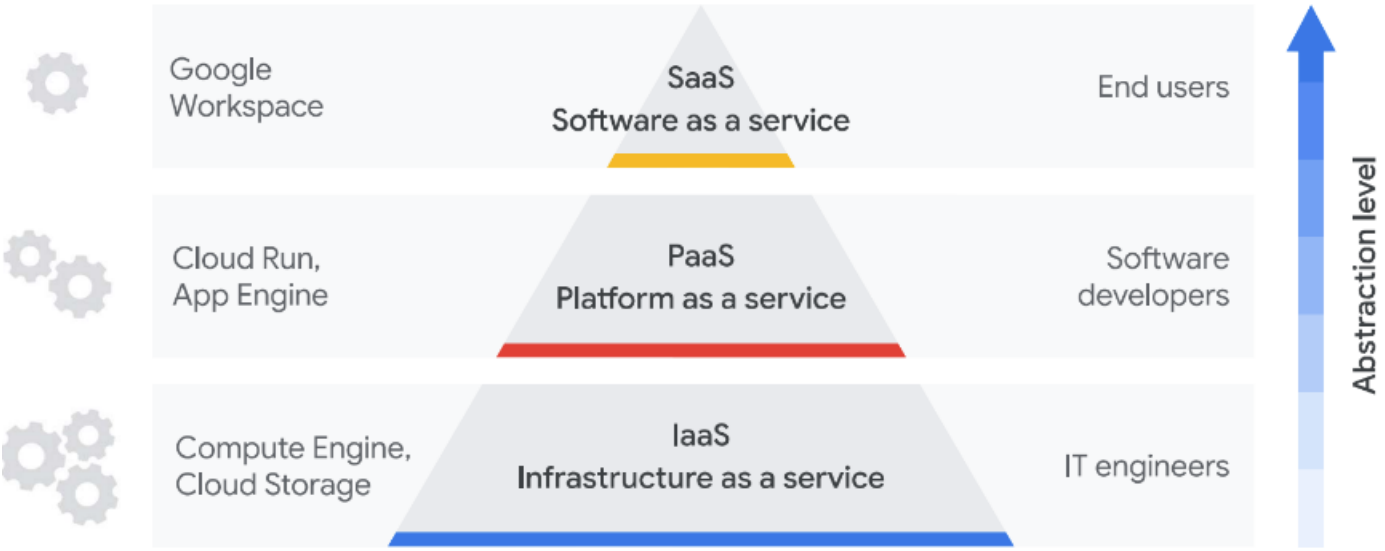
- Bandwidth - how much data a network transfer in given amount of time (Mbps, Gbps)
- Latency (lag) - amount of time it takes for data to travel from a to b (ms)

Location -> Region -> zones

Google **edge network** - entry point of network



Cloud computing models and shared responsibility



IaaS - Infrastructure as Service

- On demand availability - compute, network, storage, Databases
- Lease the resources and pay for what u use
- Reduce capex and transform opex
- Cloud providers manage infra

Benefits - Economical (pay as you go), Efficient (available when you need), Boost productivity (comp can focus on other activities), Reliable (no single point failure), Scalable

PaaS - Platform as Service

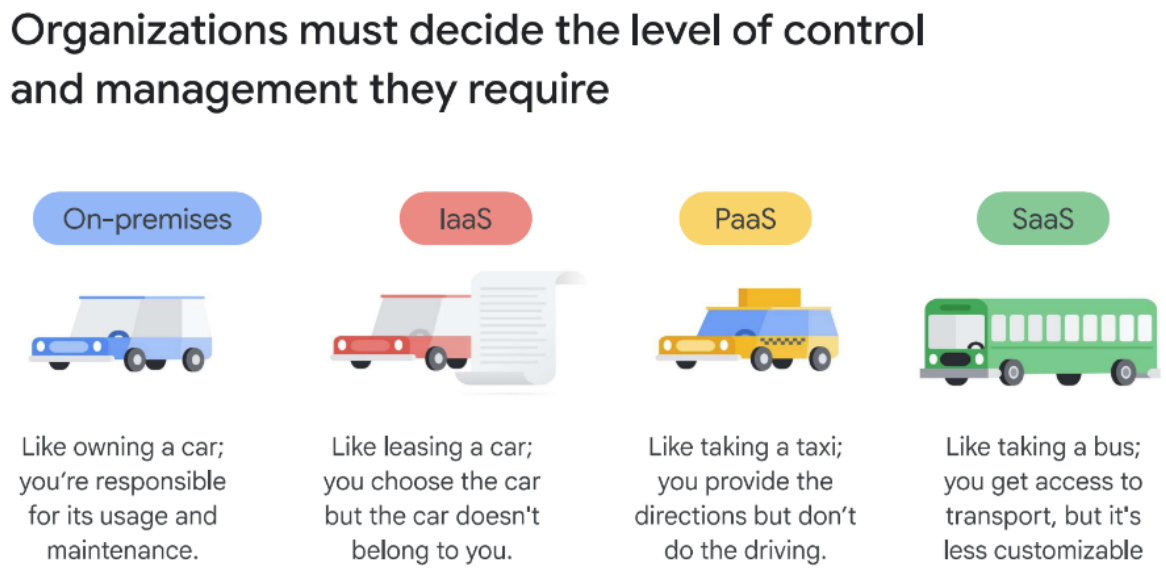
- Provide platform for developers to develop, run, and manage apps (No need to maintain infra)
- **E.g. Cloud Run, BigQuey, vertex AI, GKE, App engine**
- Benefits - Reduce develop time, scalable, Reduce infra management, Flexible

SaaS - Software as Service

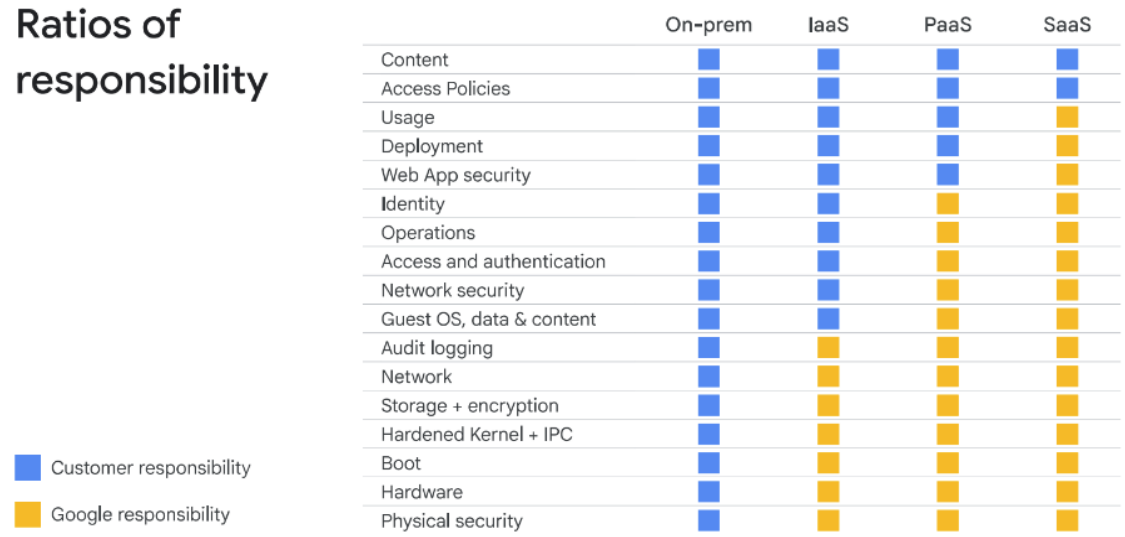
- Offer entire app as service by cloud provider through web browser
- Simply pay a subscription to use it
- Users don't need to care about underlying infra and technical things

E.g. Gmail, Gmeet, Google Drive, Google Docs, etc.

Benefits - Low maintenance, cost-effective, flexible



Shared responsibility model



Module 2 Exploring Data transformation with google cloud.

The value of data

Data is the key to unlocking the value from AI

- **Structure data** - highly organized and well defined like sheets, csv
- **Sem-structure data** - organized in hierarchy, without order or differentiation like email, HTML, XML
- **Unstructured data** - not organized like text, audio, video / considering RAW data.
- APIS can help extract value from unstructured data

- **OLTP: online transaction processing > write**
- **OLAP: online Analytical processing > read**

- **Database**: Collection of data in stored tables
- **Relational Database:** same as structure Data
- **Cloud SQL** - MySQL, PostgreSQL, SQL server (99.95% aval)
  Don't need horizontal scaling just need data store
  Local support
- **Cloud Spanner** - scalable horizontally to handle unexpected business spikes -
  high number of ips/ops per seconds (99,99% aval)
  Sharding databases, transactional consistency
  Global support

- **Non-relational database:** Same as semi-structured data
- **Firestore** - serverless, flexible, horizontally automate scalable, NOSQL, storing and syncing real time
  Data -> document -> collection
  Feature > automatic scaling, offline usage
- **Bigtable –** NOSQL big data store, large workload with low latency
  More than 1TB data, batch or Realtime process

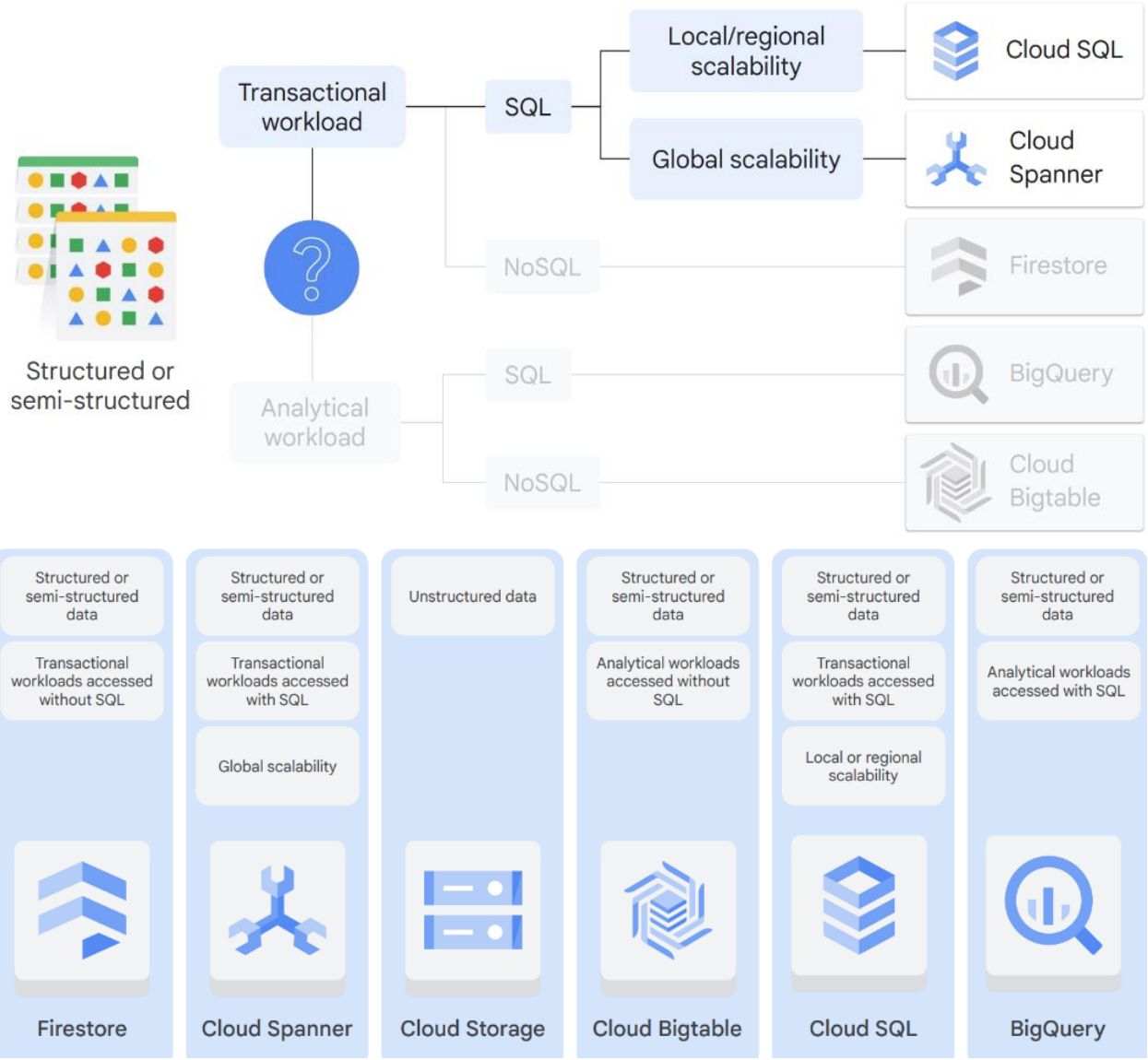
- **Data warehouse:** analysis and reporting of structure and semi-structured data
- **Big query** - fully managed data warehouse SQL.
  Storage & analytics, encrypted by default, works in multi cloud, built in ML models

- **Data Lake:** Ingest, store, explore and process raw data.
- **Cloud storage: main use unstructured Data**
  Data storage > Object storage! = file storage or block storage
  Binary form of actual data

Four class of cloud storage

- **Standard** storage - hot data (frequently access)
- **Nearline** storage - less frequent (data backup, archive - once a month)
- **Coldline** storage - infrequent (reading and writing once a 90 days)
- **Archive** storage - once a year (disaster recovery)

Important Feature: AUTOCLASS



Types of Business Data

- First party data: customer dataset collects from customer only
- Second party data: first party data from another organization such as partner or other business
- Third party data: Data collected and managed by organizations that don't directly interact with an organization, customer, or client.
  Google cloud marketplace - for third party data

Data value chain

- Data genesis - creation of a unit of data
- Data collection - extracting data / data ingestion
- Data processing - transforming data ready to derive insight from.

- Data storage - storing data after transformation
- Data analysis - analyzing the data / providing direction for business-oriented action.
- Data activation - after analysis pushing data to leaders to take decisions.

Data governance - setting internal standards for data, access permission, complying with external standards.

- **Ways to migrate or modernize database in cloud: lift and sift, managed**

Managed database migration service

- **Database Migration Service (DMS)** - to migrate data to GCP
- **DataStream** - synchronize data across databases, storage and systems.

Business intelligence tools

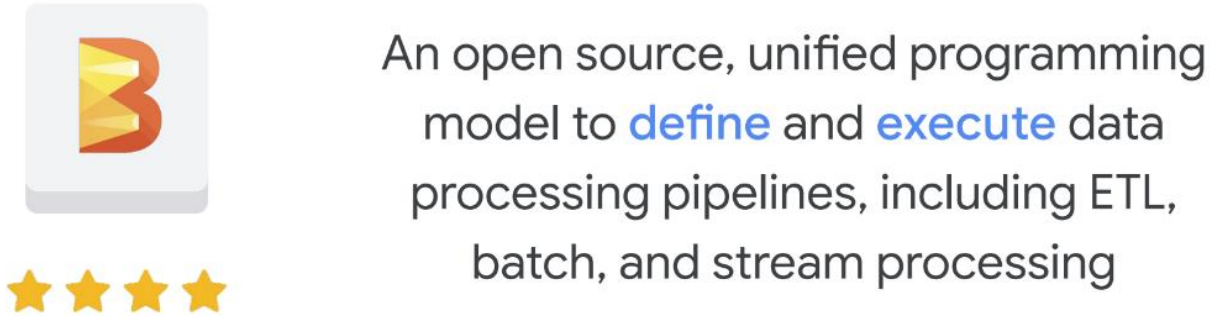
Looker - BI platform for teams to analyze and visualize data (100% web based)
lookers support big query and 60+ SQL databases

Data pipeline: Ingest raw data from different sources then move that data to destination for storage and analysis.

Streaming analytics – processing and analyzing data records continuously instead of batches.

- **Pub/sub** - for ingesting millions of events per second
  - distributed message service that receive message from different streams
- **Dataflow** (serverless)- streaming and batch data analysis with build cohesive data pipelines
  - it's referred to ETL.

Apache Beam is a popular solution for pipeline design

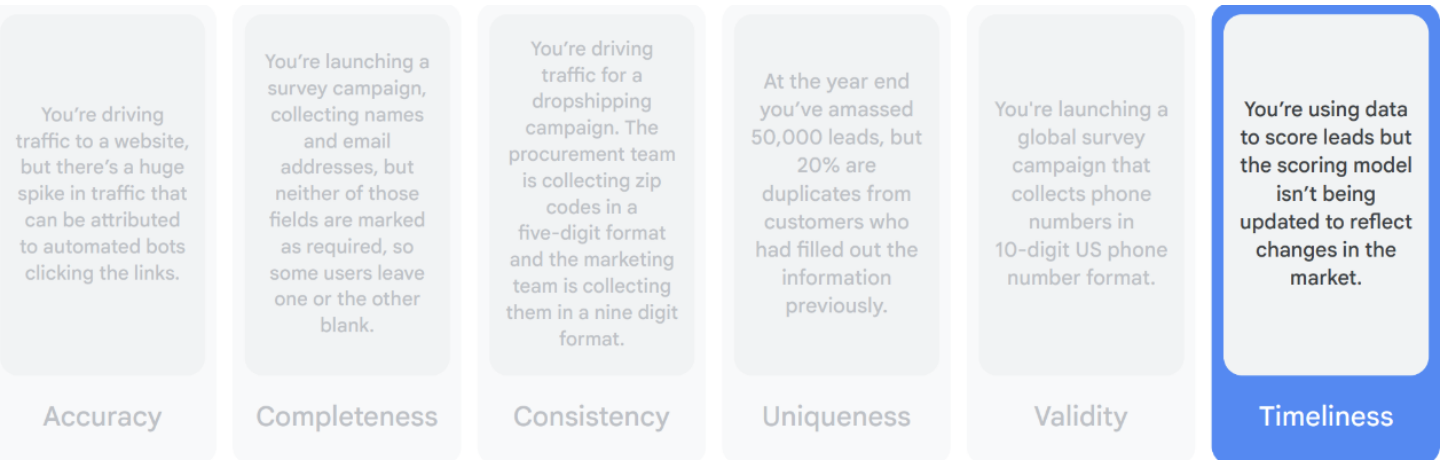


Module 3 Artificial Intelligence

Artificial Intelligence: mimic like human cognitive functions
Machine Learning: without being explicitly programmed
Gen ai: create new content, image, text, video and audio

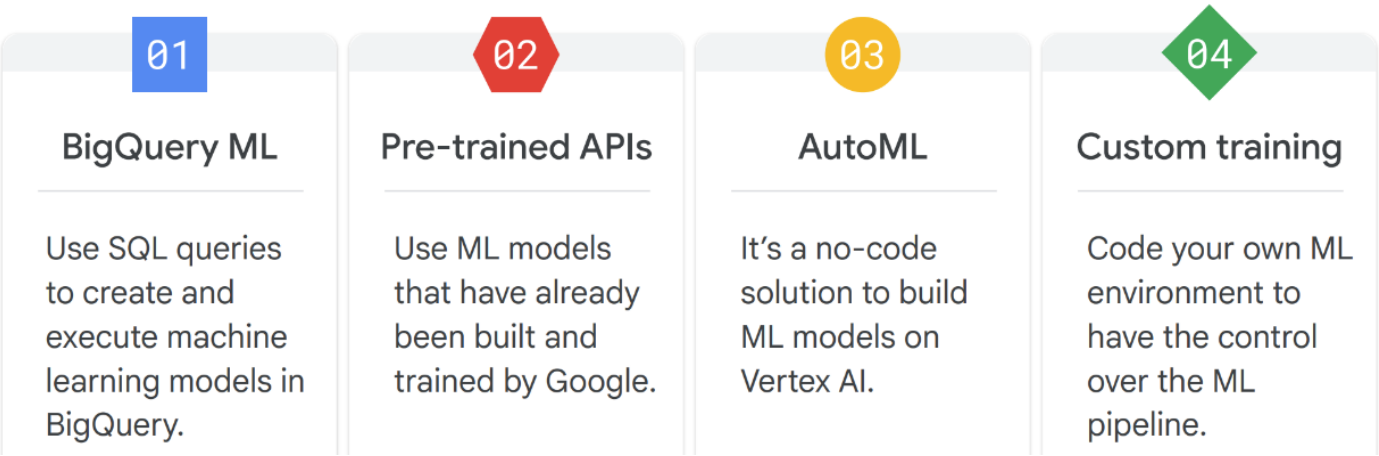
High quality data

- **Completeness** - whether all required information is present
- **Unique -** confuse by duplicate and won't understand data patterns
- **Timeliness -** no outdated data, only up to date data reflect current state of phenomenon.
- **Validity –** pre-define standards - types, format of data
- **Accuracy -** correctness of data - form, content
- **Consistency -** weather data is uniform or not

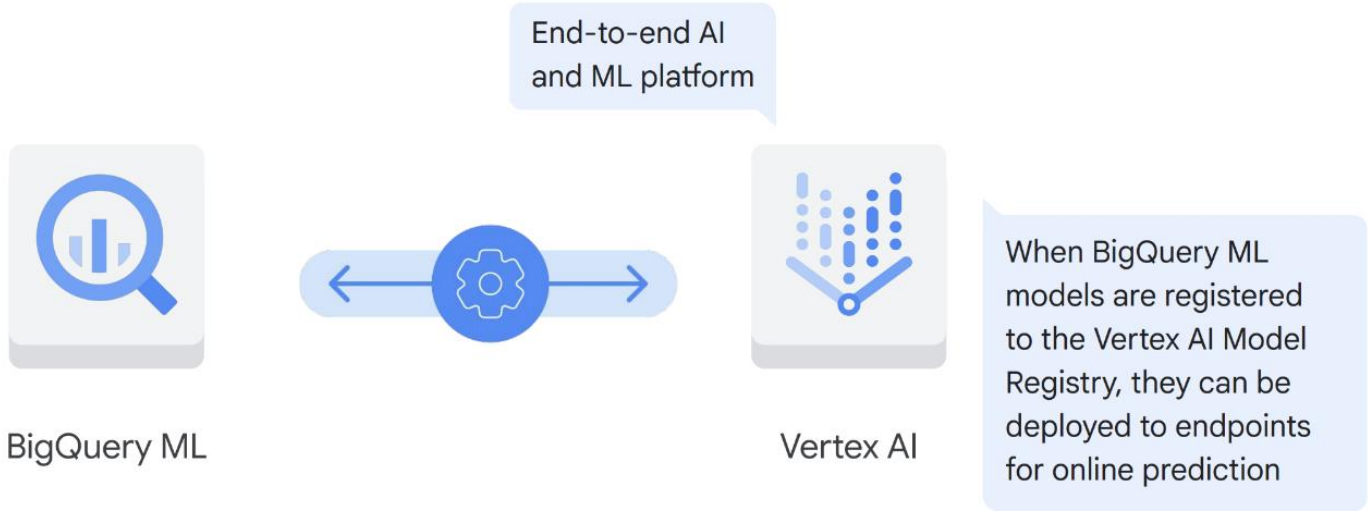


- **Responsible AI:** organization should develop their own set of responsible AI principles
- **Explainable AI:** google cloud set of tools and framework to help you understand and interpret prediction made by your own ML model.

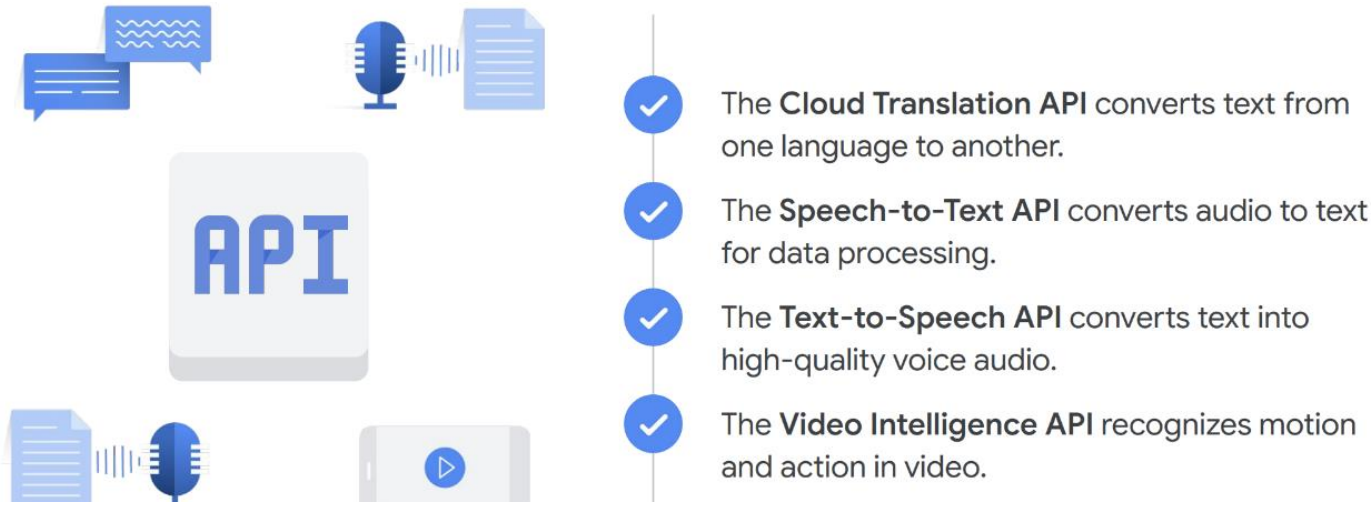
ML options on Google Cloud



BigQuery ML integrates with Vertex AI



Google's pre-trained APIs



- **Cloud vision API** > image to Assigned label Data
- **Natural language API** > sentiment text data to predefined set of categories

AutoML - no code, GUI based, vertex ai component
Auto mL Performance >> Cloud API performance

Custom training: build, manage and deploy own custom ml model in vertex ai with flexibility

TensorFlow - TPU (tensor processing unit) to accelerate ml flows speed increase more than 200 times

Google cloud AI solutions

- **Contact center ai** - for customer-care and assisting support
- **Document ai** - extract and classify information for NLP tasks
- **Discover ai:** retail product use for e-commerce site
- **Cloud talent solution:** job search and talent acquisition

Consideration when selecting google cloud AI/ML solutions

- Speed, differentiation, expertise, efforts
- Main tool: Vertex AI
- General project timing: 3 to 36 months

Scenario

Your company is looking to implement an AI solution to improve product recommendations. There's a lot of historical data on customer purchases, but the data is not well structured. In addition to data on customer purchases, your company also has access to customer demographics, product reviews, and social media data.

Considering the tradeoffs of **speed**, **differentiation**, **expertise**, and **effort**, which of the following options should your company consider?

Pre-trained APIs are quick to deploy and can be used to generate recommendations based on a variety of factors, such as customer purchase history, product popularity, and customer demographics.

AutoML can be used to build a product recommendation model based on the company's historical purchase data.

Custom training can be used to build a product recommendation model that takes into account all of the company's data, including structured data, unstructured data, and external data sources.

Module 4 Modernize Infrastructure and Applications with Google Cloud

Important cloud migration terms

Workload is a specific application, service, or capability that can be run on cloud or on-premises. Workloads include containers, databases, and virtual machines.

- **Retired -** Removing **unnecessary** workload from platform/tool/software
- **Retained – Intentionally Kept workload on prem or hybrid cloud, managed by the business**
- **Rehosted - lift and shift - the migration** of the workload **to the cloud without changing anything in the workload code or architecture.**
- **Replatform - Move and Improve - refers to migrating the workload to the cloud while making some changes in workload code or architecture.**
- **Refactored - refers to the process of changing the code of workload.**
  Like: Serverless architecture, Microservices architecture
- **Reimagined - refers to rethinking of using cloud technologies for business improvements**
  Like changing strategy from traditional way to digital or AI or ML.

Scenario 1: A multinational coffeehouse chain currently accepts orders in-person, by phone, or on their website. They want to provide customers with another way to order by building a new smartphone app. The app needs to be scalable and to function globally at low latency. Which modernization strategy should they adopt?

- Answer: Reimagine
- Explanation: The organization wants to develop a new mobile application, Instead of updating an existing product. To ensure the app functions as needed it should be developed in the cloud.

Scenario 2: A video game manufacturer distributes its products through applications on consoles and mobile devices, supported by legacy infrastructure. Consumers are complaining that the app lacks feature like cloud gaming and customizable profiles. The manufacturer wants to provide these but is struggling to integrate them into the current application and can't risk extended downtime. Which modernization strategy should they adopt?

- Answer: Refactor
- Explanation: Upgrading the existing application with these new features would be difficult due to legacy hardware limitations and the costs of downtime. Refactoring will allow them to integrate the new features into the existing application without having to risk extended downtime.

Cloud computing uses a network of remote servers to provide on-demand access to resources.

Virtualization is a form of resource optimization that lets multiple systems run on the same hardware

Virtualization relieves the pressure to bind specific computing hardware resources to specific applications

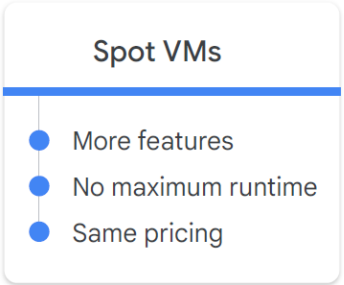
Compute engine: Google (IaaS) product that lets users create and run virtual machines on GCP.
Virtual machine instances Can be created through the:

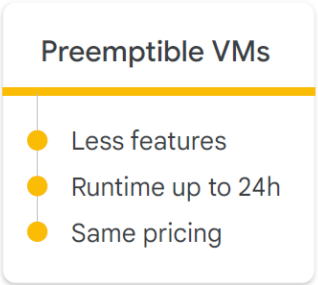
1. Google Cloud console, which is a web-based tool to manage Google Cloud projects and resources.
2. Google Cloud CLI (command-line interface) by using infrastructure automation tools such as Terraform or the Compute Engine API.

- Sustained-use discounts (SUD): VM that runs more than 25% of a month; Compute Engine applies a discount for every incremental hour of use.
- Committed-use discounts (CUD): When committing to use resources for either a 1-year or 3-year period

Preemptible and Spot VMs: Costs reduced up to 90% compared to compute engine VM.

Compute Engine might preemptively interrupt Spot VMs to reclaim the capacity at any time.





Containers: Isolated environment for running software/services and optimizing resources from one piece of hardware.

VM vs Container

- **Virtual machines:** virtualize an entire machine down to the hardware layers
- **Containers:** virtualize software layers above the operating system level

Containers start faster and use a fraction of the memory compared to booting an entire operating system.

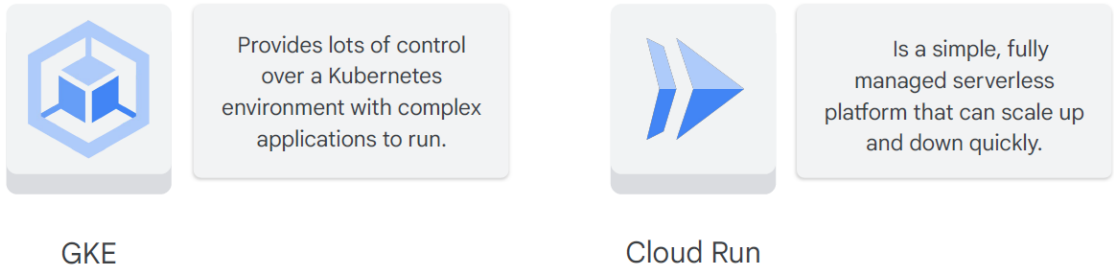
Container Management

Kubernetes: Is an open-source platform for managing containerized workloads and services.

ways **to deploy applications with GKE**

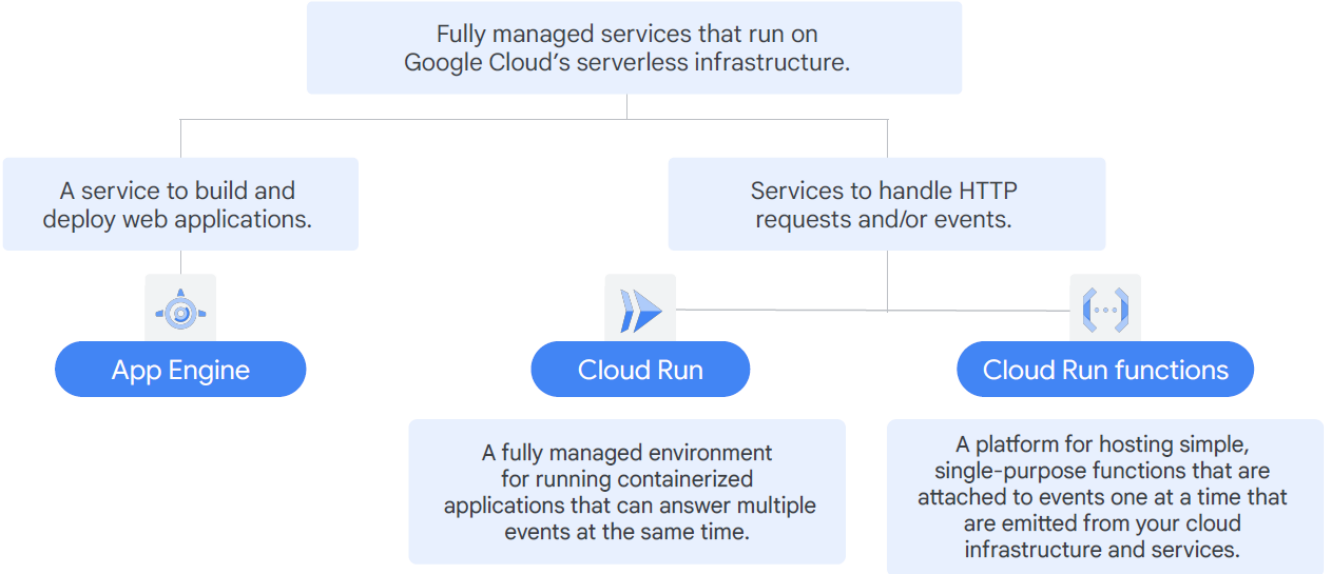
- GKE > API + Web console

Cloud Run: a fully managed and serverless platform to run a containerized app, ideal for running stateless applications.



Serverless computing - resources provided automatically provisioned in the background as needed, doesn't mean there's no server.

Google Cloud's serverless computing products

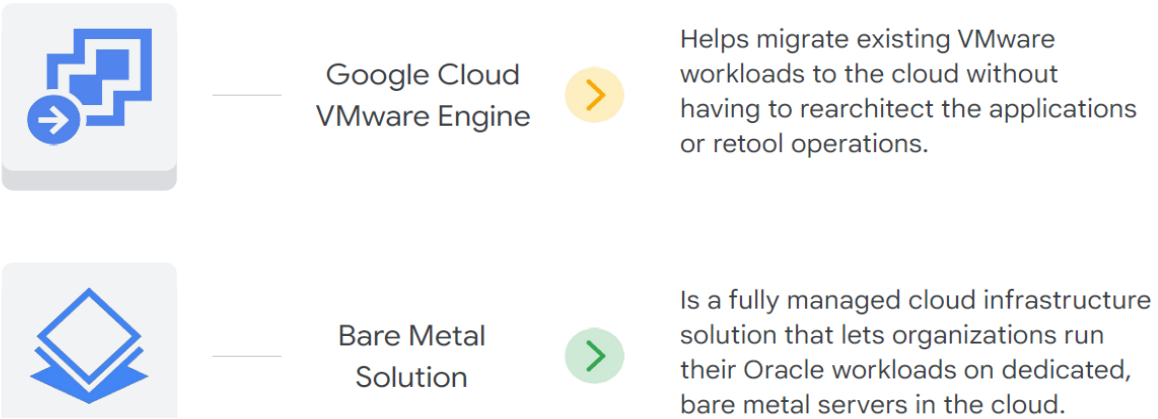


Application : A computer program or software that helps users do something.

Traditional software development or Monolithic applications: all the components of application to be developed and deployed as a single, tightly coupled unit, using a single programming language.

Rehost basic approach: Lift and shift

Solutions for rehosting specialized legacy applications



Application programming interface (APIs): set of instructions that lets different software programs communicate with each other.

Google APIs: google search API, google maps data API, translate text API
Google Apigee - API management service to operate APIs with enhanced scale, security, and automation.

Cloud Run	Compute Engine	Google Cloud VMware Engine	Apigee API Management	GKE	App Engine
Google's serverless platform that allows you to run containers without having to manage any infrastructure.	Google's virtual machine service that allows you to create and manage virtual machines in the cloud.	Helps migrate existing VMware workloads to the cloud without having to rearchitect the applications or retool operations.	Google's API management platform that allows you to secure, manage, and publish APIs.	Google's managed Kubernetes service that allows you to deploy and manage containerized applications in the cloud.	Google's fully managed platform for developing and hosting web applications.

Module 5 Trust and security with google cloud

Cloud trust and security

- **Privileged access** - grant specific users access to specific data
- **At least privilege** - advocates granting users only the access they need to do their jobs.
- **Zero-trust architecture** - no user or device can be trusted by default.

Cyber threats

- **Security by default** - security inside the apps
- **Security posture** - overall security in cloud env
- **Cyber resilience** - ability to withstand and recover quickly from attacks

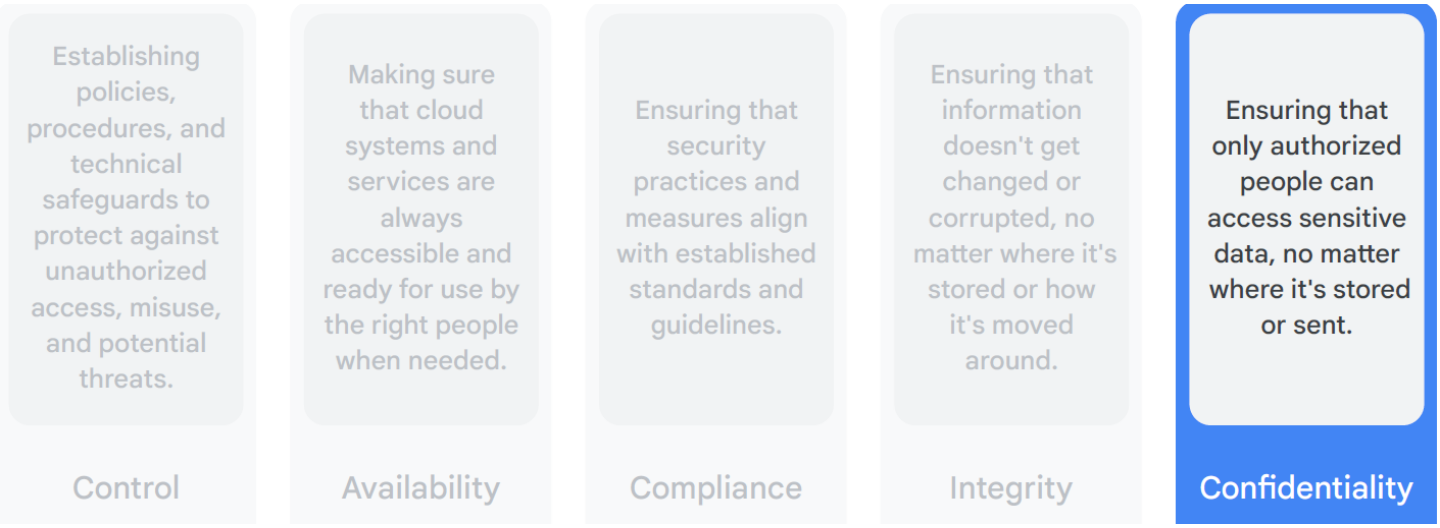
Security measures

- **Firewall** - regulates traffics based on predefine security rules
- **Encryption** - converting data into unreadable using key
- **Decryption** - using an encryption key to restore encrypted data back to its original form.

Cloud security component

3 principals of security - **CIA triangle**

- **Confidentiality** – protecting sensitive information, only authorized person access
- **Integrity** - data is not changed, ensuring data is accurate and trustworthy.
- **Availability** –maintaining uninterrupted access to resources and services for right people. / Ready to use.
- Control: manage and mitigate security risks , protect against unauthorized access.
- Compliance: industry regulations, established standards, legal requirements and organizational policies.



Cyber security threats

- **Social engineering:** Anyone within your organization can be tricked into inadvertently downloading malicious attachments, divulging passwords, or compromising sensitive data.
- **Physical damage**: safeguarding data even in the face of physical adversity

Malware, viruses aim to disrupt operations, inflict damage, or gain unauthorized access to computer systems

- **Ransomware:** ask for money, extortion, ransom is paid, blackmail
- **Vulnerable third-party systems:** third-party systems/application can transform into potential threats.
- **Configuration mishaps:** arise during the setup or configuration of resources, which exposes sensitive data and systems to unauthorized access.

Google multi-layered security strategy

- Data centers, secure storage, identity, network security, operational and device security.

Data centers

1.Zero-trust architecture -

- security at every level
- Tamper evident hardware
- Secure boot
- Hardware based encryption

2.Physical security

3.Cyber resilience

4.We measure efficiency success through the Power Usage Effectiveness (PUE) score

PUE - Power Usage Effectiveness (min PUE -> max efficiency -> less carbon)

5. scalability: quickly and seamlessly accommodate new hardware and servers

6. customization: servers and network provide us with unparalleled customization capabilities

Secure storage: based on encryption for all stage of DATA

- **Data in rest**: stored in physical device, Cloud Key management service (KMS)
- **Data in transits**: moving over networks, encryption shields
- **Data in use:** actively processed by computer, memory encryptions

AES (Advance Encryption Standard): trusted by government and business worldwide

Identity
3A - Authentication, Authorization, Auditing

- **Authentication – identify user access through unique credentials** (passwords, biometric, keys, 2FA or 2SV)
- **Authorization -** what user allowed to do in system after authentication (like admin has rights to create or delete user, standard user has rights to view list of users)
- **Auditing (**accounting) - monitoring and tracking user activities within a system by user logs and system events
- **IAM (Identity and Access Management) - security headquarter**
    - Create and manage user
    - Assigns role
    - Grant permission to resources
    - Audit and activity
    - Monitor security positions

Network Security

Beyond Corp Enterprise - for Zero trust security
Cloud VPN and Cloud Interconnect - private access for secure connection between on prem servers and google cloud

Firewall & Virtual Private Cloud (VPC)

- Firewall + VPC >> Google cloud

Shared VPC: large fence that separates each Google Cloud Project so they can work independently & safely

DDoS - Distributed denial of services for external web frameworks.
DDoS protection: cloud Armor

SecOps - Security Operations: protecting your organization's data and systems in the cloud
Help to reduce **data breaches, system outages and other security incidents**

Vulnerability management - identifying and fixing security vulnerabilities in cloud infrastructure and applications.

- **Security Command Center (SCC)** for centralized view of Security posture
- **Log Management: watchful eye on GCP env, looking out for any suspicious activity**
- **Cloud Logging** - collect and analyze your entire GCP env.
- **Incident response** - Google representatives across various domains for security.

Educating employee for best practices

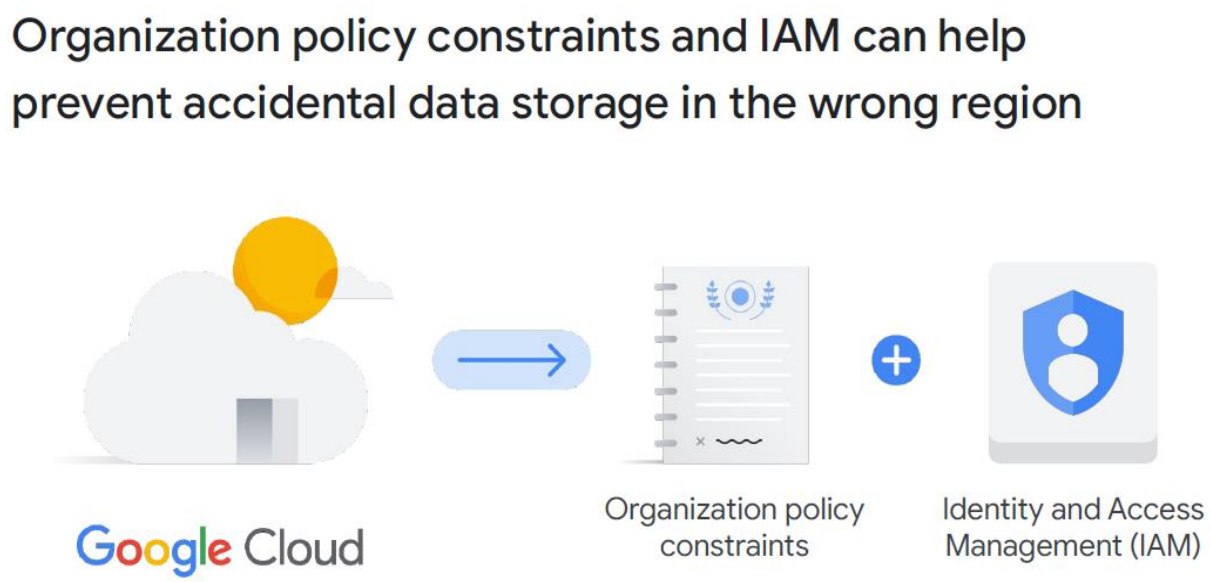
Google cloud trust principles

- You own your data, not google
- Google does not sell customer data to third parties
- Does not use data to advertise
- Customer data encrypted by default
- Guard against insider access
- No backdoor access by gov
- Privacy practices are audited against international standards

Google Cloud undergoes independent, third-party audits and certifications > EU Cloud Code of Conduct

Cloud financial governance - spend on cloud

Data sovereignty - Legal concept that data is subject to the laws and regulations of the country where it resides, GDPR (general data protectioon regulation)
Data Residency - The physical location where data is stored or processed

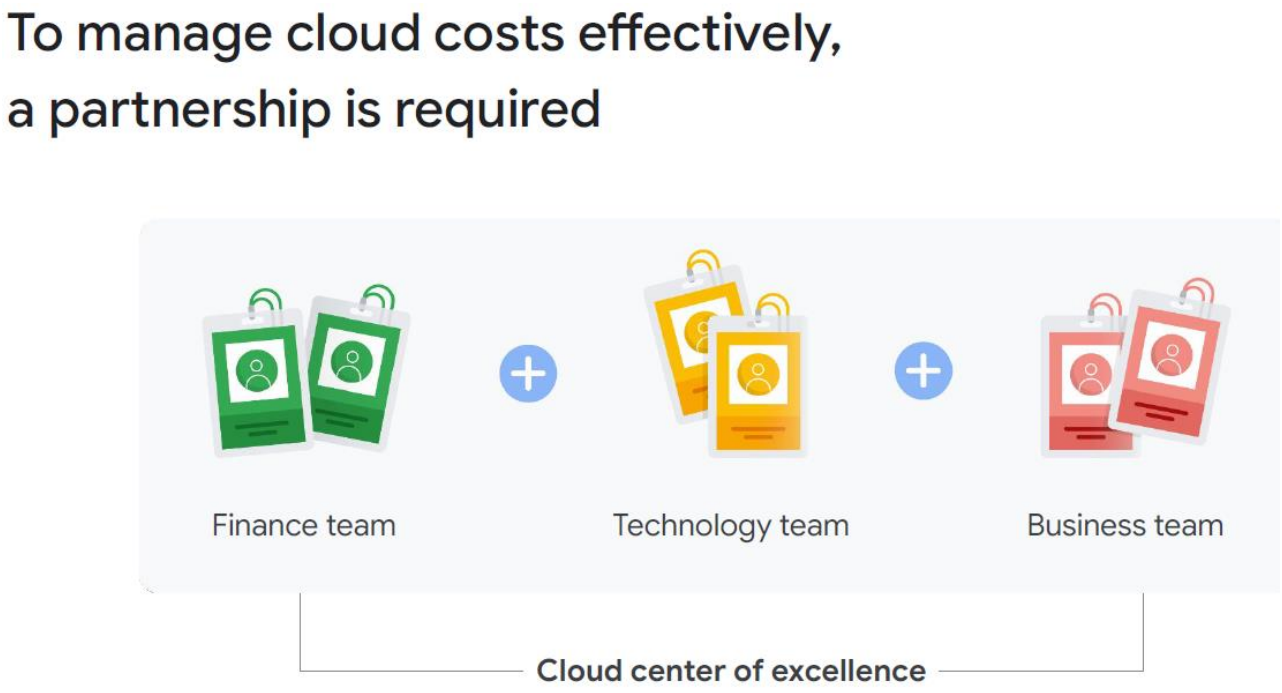


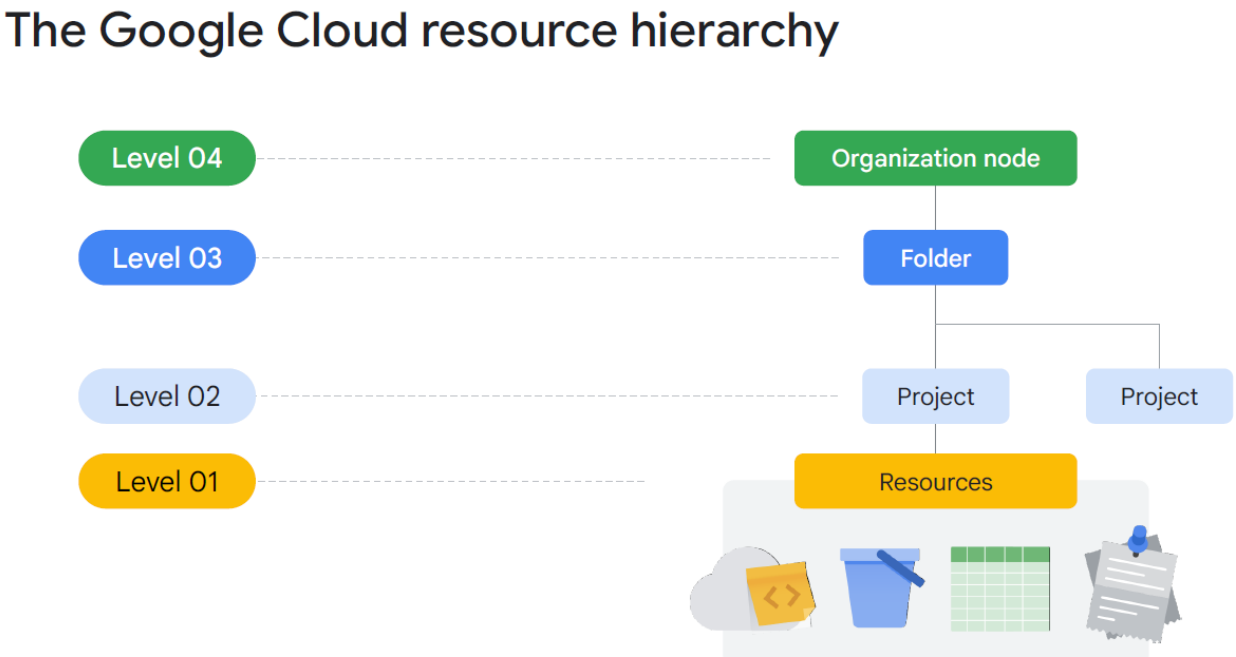
Industry and regional compliance

Compliance resource manager: Resources and tools to help achieve compliance
- discover Google's latest ISO/IEC certificates, SOC reports, and self-assessments.
- easy, on-demand access to critical compliance resources at no extra cost.

Module 6: Scaling with google cloud operation

Cloud operations - set of practices and strategies to ensure the smooth functioning, optimization and scalability of cloud-based system





- **Add policy at level 04 – organization node**
- **Policy: rulebook** that defines **who** (identity) can do **what** (roles/permissions) on a specific resource.
- **Level 03** – Folders let you assign policies to resources at the level of granularity that you choose.
- **Hierarchical Levels:** Policies are managed at four primary levels: **Organization**, **Folders**, **Projects**, and occasionally individual **Resources**.

The Power of Inheritance: Resources automatically inherit the policies of their parent containers. For example, a policy set at the folder level applies to every project inside it.

Controlling cloud consumption

- **Resource Quota limits policies** - set limit of resource that can be used by project or user
- **Budget threshold rules** - set alerts to be informed when cloud cost exceed certain threshold
- **Cloud billing reports** – reactive methods to track and understand what has already been spent.

Committed use discount (CUDs): Discounted prices in exchange for Your commitment to use a minimum level of resources for a specific term.

Fundamentals of Cloud reliability

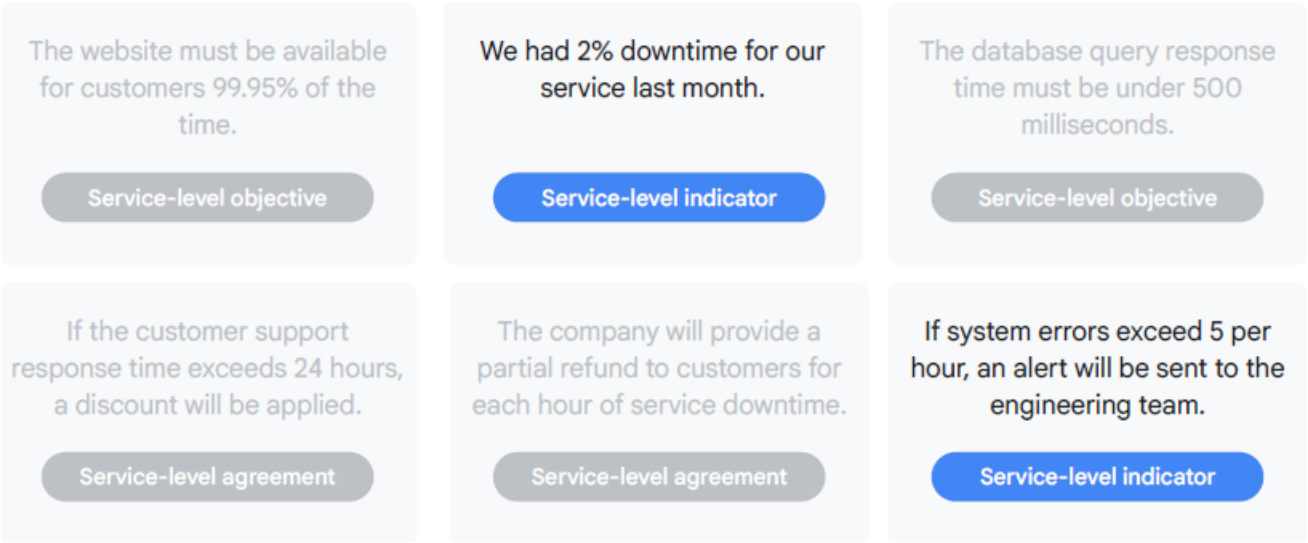
- **SRE**- Site Reliability Engineering: Ensures the reliability, availability, and efficiency of software systems and services deployed in the cloud.

Golden signal for system reliability and perf

- Latency - how long it takes for a particular part of a system to return a result.
- Traffic - how many request
- Saturation - capacity of system
- Errors - measures system failures

3 concepts of cloud reliability

- **SLIs (Service Level Indicators)**- measurement that show how well a system or service performing (response time, error)
- **SLOs (Service Level Objectives)** - level of reliability or performance that we want to achieve.
- **SLAs (Service Level Agreements)** - agreement b/w cloud providers and client
    - agreed-upon SLOs, performance metrics, uptime guarantees, and any penalties or remedies if the provider fails to meet those commitments.



Designing resilient infrastructure and processes

HA vs. DR: High Availability (uptime during failures) vs. **Disaster Recovery** (restoration after disaster).

- High availability refers to the ability of a system to remain operational and accessible for users even if hardware or software failures occur.
- Disaster recovery refers to the process of restoring a system to a functional state after a major disruption or disaster.
- **Redundancy:** Duplicating critical components or resources to provide backup alternatives.
- **Replication:** Creating multiple copies of data or services and distributing them across different servers or locations.
- **Multi-Region:** offer multiple regions or data center locations across different geographic areas.
- **Scalable infrastructure:** varying workloads and accommodate increased demand without compromising performance
- **Backups:**
  - If data loss, hardware failures, or cyber-attacks occur, organizations can restore their systems to a previous state.
  - Backups should be stored in geographically separate locations.

Validation: Constant **monitoring and testing** to ensure recovery plans actually work.

Modernizing operation/observability using google cloud

On-premises: physical check

Google cloud: observability tools (cloud monitoring, trace, profiler, so on)

Observability - collecting, analyzing, visualizing data from various sources
Google Cloud Observability - Cloud monitoring, cloud logging, cloud trace, cloud profiler, error reporting

- **Cloud Monitoring:** Tracks **metrics** (performance data) and triggers **alerts** based on specific criteria.
- **Cloud Logging:** A central repository for all system and application **logs** used for troubleshooting.
- **Cloud Trace:** Analyzes **latency** to find performance bottlenecks in distributed applications.
- **Cloud Profiler:** Monitors **resource usage** (CPU/Memory) to optimize application efficiency.
- **Error Reporting:** Aggregates and analyzes **crashes** and exceptions in real-time.

Google cloud Customer care -

- **Basic support** - free for all ()
- **Standard support** - for development, tech support, troubleshoot, cloud support Api
- **Enhanced support** – for production, fast response, 24/7, selection of language, third party issues
- **Premium support** - for enterprise with critical workloads, higher management, google skill training

The life of a support case

- **Case Creation:** create case a through tech support editor
- **Case triage:** review case creation, Request additional information or clarification
- **Case assignment:** If customer care representatives could not resolve then it will be assigned to support engineer.
- **Troubleshooting and investigation**: Analyze the provided information, review system logs, conduct diagnostic tests, and collaborate with other internal teams or experts.
- **Communication and updates:** Customer Care team maintains regular communication with the customer. provide updates on the progress, share findings, and request additional information or actions from the customer
- **Escalation:** flagging process breaks or for the rare occasion that a case is stuck because a customer and the Customer Care team aren't fully in sync. Ensures that the case is assigned to the right resources as quickly as possible.
- **Resolution and mitigation:** resolving the issue or providing a mitigation plan.
- **Validation and testing:** perform specific task and provide feedback
- **Case closure:** issue resolved.

Sustainability

Google's data centers were the first to achieve ISO 14001 certification.
Founding decade – carbon neutral
Second decade – 100% renewable energy.
By 2030 – carbon free

enhance its environmental performance Through improving resource efficiency and reducing waste.